2023 Video ReviewStream ~ Active Inference Institute

ReviewStream | Active Inference Institute | 1:03:56

hello and welcome it is December 22nd
2023 and we're in the 2023 active
inference video review stream so welcome
to the active inference Institute we're
a participatory online Institute that is
communicating learning and practicing
applied active inference this is
recorded in an archiv live stream so
please provide us with Fe feedback so we
can approve our work all backgrounds and
perspectives are welcome and we'll
follow video etiquette for live
streams head over to active inference
dorg to learn more about live stream
projects and other projects and
groups today's going to be a fun stream
so thank you Dean for
joining the goals on the slide are
reflective learning discuss deriving the
23 set of posterior probabilities and
look towards the 24 anticipatory set so
maybe by way of introduction could you
go a little bit into that
Dean well we we always find ourselves at
this point in the year thinking well
okay
well have we closed the loop and and are
we opening up another one and
that's that's something that uh I kind
of look forward to because it's not just
a sort of a moment in the calendar it's
also a question of whether or not we see
what we've done in the in the in the
previous lap around the sun we see
ourselves in a different way and I think

we well I won't speak for you Daniel but
I certainly do every time we we reach
these uh moments uh I I I kind of look
forward to looking
back yeah I as well so big questions
what was our stream of attention in 200
23 we have courses mechanics and guest
dreams Galore in the so very dark
Imaginarium so what does that mean for
you
first um well there was there was a
certain presenter at our Symposium this
year that her her talk was about a a
dark Imaginarium and and you know what
would that be like what would that what
would that space look like and how would
you quantify it and so I think one of
the things that just the sheer variety
and the ways that information was shared
this year um spoke to the fact that
there is no one way of seeing things or
presenting things or making things
available and so that's why I meant by
so very dark
imaginary okay nice and then where and
how will we stream in
2024 more new series voices formats and
languages continuing and developing the
forms we know and love more podcast and
online formats and revisiting materials
with the actm journal what else would
you add or maybe let's leave that for
later let's jump into the main um
triptic let's jump in there and then if
you're watching live let's return at the
end the mo most of the stream is going
to be going through each of these series
just kind of one slide per series and

reviewing the 141 videos that were prod
this year and uh we'll return at the end
to where we want to go in 2024 and
whoever's there will be included in
23 um yeah there were 141 videos overall
here's the sections that we're going to
go through and the products here are
placed into the inbox of the active
inference Journal where we have
utilities that are transferable outside
of our Corpus and then also appi to our
Corpus it results in the active Journal
repo and so that's where we work on the
transcription translation so a story for
another day but thank you to everybody
who works on that side of things so any
comments before we jump into the
series lots of different ways of
creating a paa even though it from a
from a a formatting and structuring uh
standpoint we
all we all never know I don't think at
the beginning of of our next cycle
exactly what's going to end up being in
the in the big pot of things and that's
one of the other reasons why we do this
thing at the end of the year just so we
can uh kind of see what the difference
is between what we thought we were going
to be and what we actually turned out
like all right let's do it so we had
four
roundtables these are Institute scale
upd
meetings there are four of them they
were informative we hope and Dean you
wrote keeping the pie and pizza
explain yeah we just the fact that the

round table seems like an an appropriate
metaphor the um the whole idea of making
sure that how we communicate with the
world outside of the Institute I think
really matters and I think you guys do
an amazing job of making sure that
anything that you
believe um will make it easier for
people who may not current ly be engaged
with us to find out who we are and maybe
get a better sense of what we're about
um you guys work on that really hard and
appreciate the fact that um not
everybody gets what the active inference
stitute is but you guys are certainly
pumping out as much um dialogue and and
information on that as you can thank you
I was thinking of the joke like oh
please cut the updates into four pieces
I'm not sure I could could eat
eight all right into the main content so
leading off we have the course with
Chris Fields physics is information
processing and with Andre rir as the
assistant this was an amazing series
there were six lectures that were inter
woven with discussions and
interactions Dean what can you say
well first of all thank you to to the
the three primary um people who spent it
again just a an immense amount of time
preparing for this and making it
available to people free of charge I
think it was one of those moments where
the active inference Institute could
have and Chris and yourself and Andrew
could have easily questioned whether or
not there wasn't some value that you

couldn't have been able to draw from
this but I think the fact that you put
it into the common space and gave
everybody a chance to be able to have a
look at this uh just speaks to the fact
that um there's good people who have
lots of good information and open
themselves up and make themselves
available so thanks to three of you and
then everybody else that joined in at
various
points yeah and really it just
represented
an already consolidation of some form of
the quantum free energy principle that
Chris fields at all have been working on
only in the last several years so that
was already cool to see and the second
course sounds like a meal already the
second course um of the
year was the active inference for the
social sciences constructing cultural
landscapes so this was a great
collaborative work
with AEL gwin and Caro Ben White Mal
abasan and lorenzza ganela myself and
AEL kind of had the vision for this
course and it was co-organized with cyos
research each of the people listed had a
topic so we kind of
roughly blocked it out that way with
ail's vision and collaborating on how we
wanted
to consolidate and bring together some
of the threads of discussion around
active inference in Social
systems we had AVL with an introduction
Ben basics of acadm an incredible

lecture I did a section on Collective
Behavior Lena on semitics and semantics
Ma on Norms and avel with constraints so
this was really fun and we hope that the
people who joined the discussion enjoyed
interacting with it it just felt like
the opening of a lot of increased
interest in active inference for social
systems and just the beginning of the
discussion on it but really cool to work
with multiple co- teers and do this
second course using some of the
templates and lessons that we were
learning from the chrisfield
side all
right we had the third PL active
symposium
enacting ecosystems of shared
intelligence and this was really fun
there were a whole team of co-organizers
so thanks to everybody who did that and
all the guests
were listed on the side there were two
intervals the program still up we had a
great Round
Table what do you think about this
Symposium or what would you
add I think you must have had to drink a
lot of coffee because
you you're kind of in the thick of this
uh of this weather system for a long
time I think it was amazing I mean the
the the support from our presenters the
support again from the people who made
sure that everything sort of came off
with a with a reasonable amount of
consistency and Co coherency um again if
you want to if you want to be able to

get a real clear sense of how much
change there's been going on in the last
few years and how Dynamic the
understandings around active inference
are simply open up one of these
symposiums and just get a get a sense of
how how much the spectrum of research is
continuing and
evolving
yeah lot to say on that too but it was a
fun topic and a really
memorable
Symposium with an application focus and
also I remember Pablo presenting the
first AC game and us being in the the
digi cyber physical cognitive ACM game
space and a lot of other memorable
moments there there was the dark
Imaginarium there which is still one of
my one of the
greatest fantastic moments to listen to
that that was awesome
yes all right well
Darius did incredible work in the last
several weeks with bringing out the
first five episodes of the insights
Series so we had Carl friston ma operas
John Veri iness hippolito and Marilyn
stera just in the last several days
honestly each of these were incredible I
learned so much from returning guests
and from new guests and it's really fun
Darius thank you for stepping up and for
doing a lot of the prep
and having these conversations that that
speak for themsel so it's really fun to
see any other comments on
that Darius says I don't know how what

his background is but he's got a face
and a voice for podcasts for
sure
nice all right into the series we go
morph Stream So Sarah Hamburg over here
proposed as part of the scientific
Advisory Board and just all around
active friend proposed this neuromorphic
plus active
inference combination stream and so far
there have been two morph streams shown
here and it's really been cool and it
will be fun to see where this goes as
the neuromorphic and unconventional
Computing spaces develop and as active
inference is used in the measurements
and modeling and design of
unconventional Computing and synthetic
intelligence this is going to be a
really exciting stream so thank you
Sarah for leading these discussions and
reaching out to people and everything
that's something weird there morphic
what is is that a neck beard that you're
sporting in that picture like what's did
you morph into something what happened
there not
sure is this been is this modified is
this is this I'm not sure oh wait a
second I was like wait a second you took
this really
seriously I barely know what to say next
hilarious well played sir
um we had orc stream with shre Jane on
Communications and secrets in
organizations and with anamarie Swan on
the ecological organizations framework
and a really interesting and relevant

topic active inference for and with and
by and in
organizations and in ecosystems so it's
really fun to have these conversations
people who are in different places
coming to active inference and having
them share what they're working on and
how that relates to organizations and I
know this will also be a fun Direction
next year anything
that all right math
stream we had three of them we had
Christian Bodner on topological deep
learning graphs complex and sheaves Shan
tol on active inference and string
diagrams with Ali and Vincent Wang masan
on Constructor Theory as process Theory
with the very memorable chalky Dusty
board and absorbent
jacket definitely a Time series and a
work of art and
motion but overall these three not to
say that math didn't come up in other
streams it did but just kind of looking
at these three it's kind of cool to see
the discret and the
topological and the categorical the
applied category theory side with Shan
tol that was also there in live stream
54 which we'll come back to but those
are some developments that are already
being applied to active inference and
some that are just kind of one step away
so a few of those discussions started to
forge some paths that are really
exciting for the math of active
inference yeah when it gets to the
sophistication that these these folks

are able to bring to it I just had to
tap out there were I mean I can get up
to category Theory and and at least fake
my way through some of it but I was I
was amazed at the
uh just the confidence that some people
that are are bringing to this this whole
idea of where where it can it go yeah
it's
incredible all right model stream so
these are streams that focus on the
generative models and do some kind of
Step through with models so we had model
stream 7 the second part in the pmdp
series here with Carl Adam Connor and
yakob fun discussion and really applied
there model stream 8 with Tom ringstrom
on reward is not necessary part of a
really fun Technical and broader
discussion on what's the role of reward
what's the role of goals surprise all
these different cognitive phenomena
which ones entail one another which ones
are super sets of one another and this
kind of model uh without reward that Tom
proposed and then model stream n with
aswin Paul on efficient computation and
active inference also a symposium guest
but aswin and at all have been doing
some really awesome work with a
computational efficiency and
kind of tractability of applying these
models so that's really cool to
see all right textbook groups so there
were three cohorts that had some
activity this year there was cohort 2
which finished the second part of the
book second chapters 6 through 10 in the

beginning of this year cohort 3 which
was entirely within this year both
halves and cohort 4
uh or
sorry someone finished it half someone
started half one of them was complete
cohort 4 was the one I guess that was
completed in 2023 but the dates really
don't matter we did a bunch of textbook
groups with this
2022 par at all textbook
and it was really
inspiring there are so many ambitious
and excited Learners who come 0 1 3 25
times and people engage in so many
different ways with the synchronous and
the
asynchronous and people who just want to
listen have the
recordings so that has started to
develop really something special in
being there and just appreciating
everybody what they bring to this
book and also
Ali in particular for really bringing a
complimentary set
of very exacting accurate understandings
of what is in the textbook and also very
broad understandings of what's not in
the
textbook we had the book streams so Ali
and I did some book streams on the
2022
textbook also with Tyler and blue we did
a series on Bon hz's book governing
continuous transformation reframing the
strategy governance conversation so this
was pretty fun to go chapter by chapter

with Tyler and blue and talk about what
does free energy principle and active
inference mean for organizations and
talk about just a lot of cool topics and
have B John join for some of the
discussions too yeah any other comments
on the book
streams just to just to thank you to the
uh authors when they were able to uh to
join us because I think that again it
brought the book To
Life yeah thas Ali because
when whenever Ally comes and and joins
us you can be sure that he's going to
he's going to bring his aame he never
comes without being the most prepared
guy in the room so that's saying
something because Daniel likes to be
well prepared
so very true very true but these were
fun okay now we'll go through a couple
of slides with the guest stream so this
is a really broad
category
um it started out in January with Adam
peace talking about Sumo and neuros
symbol IC text Ali Resa modish talking
about the 18 subtypes of surprise avel
sharing some work on the physics of
creation then a very fun and memorable
conversation with Matt Bowski and Jordan
Hall J Benjamin falland days talked
about gibsonian ecological psychology
and resonance Baba
Brinkman shared a lot about the artistic
process and did some freestyle rap
played music on the stream which was
really fun Max Berg talked about

rumination shui at all talked about
richness and ineffability and this was a
philosophy with some technical
details uh van Visa talked about
language models and Consciousness Elliot
Murphy and Steven pend DOI had uh at
times contentious discussion on language
models and Chomsky that was quite an
event and ju Diego Bata and Zacharia
jebber had a conversation shown here
with Dean in hat and Ali on time
Consciousness and computational
phenomenology so any of those that you
want to comment on well well there's a
few I thought that I thought that Jordan
and Matthews um you call it a dialogue
they I think they were trying to
understand one one another it was uh
there was a bit of a c circling going on
there but I think they they kind of left
with a better understanding of each each
person's position Baba's was I never
anticipated what where that was going
and I just I loved it by the end I was
just like wow there's there's some
people who who who absolutely can uh can
see things in ways that really open up
your your idea of what a possibility is
and for me one of the ones that I really
enjoyed was um participating in the on
the time Consciousness um question I
found that any time that people will
place an emphasis on change and dynamic
change to go along with some of the more
stable measures that that kicks out um I
want to sort of figure out what they're
pointing at and so I thought they did a
really good job of um trying to explain

where where they were coming with that I
actually made connections between the
stuff that they were talking about and
the um stuff that Shanah doson talked
about in the Symposium with the dark
Imaginarium again you have to kind of
see them Side by each but where they're
talking about how we divide time into
objective time and subjective time and
how you're in the moment and how you can
see yourself as being having to be
outside the moment exactly at the same
moment and is that a contradiction or is
that complimentary I just thought that
um there was a lot of that possibility
this year but again you don't
necessarily see people coming using the
same language or even using the same
Quantum frames and yet they're still
talking about the same thing if you can
put the prepare and the measure up on an
equal footing and let those let those
two things work in concert uh it seems
to draw out of people this oh my
goodness have we thought about it in
this way so that was what I saw from
this first segment of the three parts of
the guest streams this year awesome all
right guest stream part two so Tom froze
on eruption
Theory really cool part of a bigger
discussion about Free Will agency all
those topics Bobby aarian coming back
for
15.3 on the theological stance and
building that road to Omega uh suchia
and saigo on category Theory and
Consciousness and integrated information

Theory this was really fun it's always
exciting to see category Theory drawn
out in real time and what people show as
they construct and Trace so that was
really fun and I think it also
triangulated with a lot of discuss that
we had been having about information
information Theory integrated
information Theory all this discussion
about
Consciousness different kinds of models
um Maxwell and Ma joining to share one
of the Crossovers with Quantum
information Fe and Consciousness on the
inner screen model with with some other
guests and a super fun convers
ation Denise Hol active inference AI in
the spatial web symbolizing a year of
excitement around the spatial web and
various other developments in the active
inference EOS system and how that was in
June
2023 how that is in December
2023 Sam bini leche on predicting
reflame River reflecting framework for
dual process Theory more large language
models and cognitive science and action
Adam saffron and Arthur gilani on deep
canals was a fun discussion with a lot
of mapping and territories in the brain
Megan Peters and Nora Bradford having a
very interesting conversation about
Consciousness research and how it's
communicated and carried out so very
complimentary to a lot of the other
discussions
ongoing beatini and C Yuna discussed
selfless mind unlimited bodies bringing

together some of the basian
phenomenology with meditation and
different ways of
knowing salvatory causal inference via
predictive
coding the uh elade and Dell G gestring
on the backprop free 4D ant-based
search pretty fun representing kind of
like The Cutting Edge in ant algorithms
which is always a fun opportunity to
talk about ants and
active and Alexi to chinsky who led this
discussion of one of his recent papers a
case for Chaos Theory inclusion and
neuro psychoanalytic modeling and just
brought together this panel and so that
was very memorable because there are
different voices just it was like the
paper being released and just having a
participant in our learning groups and
part of our live stream series and just
fostering the discussion just seeing how
rich and how how people see these works
that are coming out month by month so
any comments on any of these
streams no in the middle of the year I
was being a carpenter and I was also
trying to make sure that I could say
something coherent about category Theory
and Chris Fields stuff so I wasn't I
wasn't present for most of these fair
enough all right then the last third of
the guest streams so Andres Gomez
emilson and Chris Percy on the
electromagnetic field topology
Consciousness so this is the kind of qri
Consciousness qualia research very fun
and also really

provocative Pang and fornito on
geometric constraints in human brain
function slow traveling waves in the
brain uh Gregory Sergent on agency with
structured latent spaces on the math and
phenomenology side Andy Keller on neural
structure and artificial neural
networks Carl friston Chung and Schwarz
joined for a very interesting discussion
on working with Gerald Edelman who won
the Noel prize and was a very uh
influential researcher in the
neurosciences and hearing from each of
these three fellows their experiences
working with
Edelman was
incredible and it was
really
historic and gave a lot of context on
the theory and the
people
and I hope we can do more they were
impressionable young men that's what
they
revealed they shared about that time in
their lives and that time in in how it
was and and uh I I I look forward to
more of this working with format so that
was pretty fun um Kristen Whittle
Reflections and implications of a naive
attitude to
architecture very beautiful presentation
and and far-reaching
and and Majestic IC and touching upon a
lot of aesthetic comments and um very
poetically addressing questions about
applying active inference in practice
not applying it in theory in

practice um David Tucket conviction
narrative Theory narrative and active
ever green
topic Ramen on myth of objectivity and
the origin of symbols going a little bit
deeper from The Narrative into the
symbolic Michael Carl uh great
textbook group participant presenting
his research on deep temporal models of
the translation process which led to
some very interesting discussions about
the ey moving and the heat the keyboard
typing in the translation process Mel
pakari gave a very philosophical
presentation on incorporating
variational free energy models into
mechanisms the case of predictive
processing under the free energy
principle and I'm I'm a big fan of this
presentation and paper and here's why
when people are giving mechanistic
accounts for different physical material
systems the the idea that um one would
doubt the relevance of a material
physical flow is almost like heresy like
of course the windmill works because of
the flow of air or however one wants to
say it but the flow of gas gasoline the
engine things like that and yet here we
are talking about free energy flows so
framed in many ways we have this
question that sometimes we take a stance
on or not is what is the status of free
energy flows is it just an accounting
device that's just epip phenomenal or
does it have a kind of causal Force
within an explanatory account just like
the flow of water downhill power

this energy generation could it be said
for some or which systems that flows a
variational free energy play an actual
causal explanatory
account and in what situations can we
only pull back to say well this is
playing a role in my surveying of this
situation but not necessarily playing a
causal role in that territory but yes it
plays a computational role on my map
that's the kind of instrumentalist
stance so this was very deep work that
validated and connected those
discussions that are ongoing in the
active inference bace to broader threads
in history of
science um Elliot Murphy talked about
Rose a neurocomputational architecture
for syntax maybe we'll come back to that
one Adrien Bon with Susan Hasty on the
physics of Life how to predict Evolution
a fun and far- ranging
conversation Phoebe clet and Dam Simpson
on towards basing World models was
really cool talking about some uh
current trends in thermodynamic Ai and
causal World models and then the most
recent Gest stream with Andreas kada on
the logic of ntqr evaluation of noisy
judges which was also very memorable in
probably multiple ways so what which one
of these guest streams do you have any
comments
on well one of them one I want to I want
to just shout out to Andreas for being
so passionate and being so confirmative
he I I just love the way that when you
and yob were interacting with him he was

just like yeah you get it you get it and
it was was just it was really cool to
see somebody who
obviously has capacity that are amazing
but actually when he can have a
conversation and an interaction with
people who get what he's obviously you
know invested a lot of his his time and
his passions into when he can have that
conversation with other people
who understand and can converse with him
on that that was just that was I thought
that was amazing
so yeah it was very fun all
right we're gonna do one slide for each
of the five paper live streams and then
that will be the end so we'll do five
more slides one for each of these live
streams then look at the live chat see
any questions or topics and kind of talk
about Reflections where we go okay so
earlier in the year live stream 52
geometric methods for sampling
optimization inference and adaptive
agent by and dasta at
all and I prepared the zero and had a
one and a two with Lance this is some of
the clearest most
updated most concise descriptions of
some of
the signal processing and control
theoretic and basing mechanic aspects of
active inference and the free energy
principle these were super informative
ative the kind of motif or image that
stays with me is this figure with the
ball rolling downhill which we talk
about all the time path the least action

free energy landscape most likely thing
happening Baye optimal ball rolling down
the hill and then it's falling downhill
here and it could be like through air or
through water or through honey and so
different acceleration Dynamics on that
bowl and then different ways that you
can use the statistics of the
acceleration to jump ahead and do an
accelerated optimization by jumping
ahead not too much or not too little so
kind of using the computational
resources adaptively so that you
sample and this was just a really great
discussion so big thanks to Lance for
joining timing acceleration in
Geometry add water and you'll have a
fantastic
Brew try at
home
yeah okay
53 one of probably the most memorable
moments of our life Dean to be real yeah
yeah um with uh the set of papers snakes
and ladders in paleoanthropology and to
copy or not to copy
with Walker manrique and friston and
Dean and I had a
multimon Foothill Journey from the base
camp to the do one and Pierce the Veil
with a DOT zero in the side by side by
side by side and we had the dot one and
the dot two with Michael and Hector Carl
joining so what can you say about this
series well first of all
our guests were amazing Carl Michael and
Hector Inc again people who are really
really invested in their craft and

really really are
not ready to sort of just be herded into
a certain way of thinking being uh
adopters now of what I think Michael and
Hector see as something that can really
open up spaces which is act an active
inference lens on their work in compared
to psychology
and anthropology and
archaeology I want to shout out right
now that I think one of the one of the
greatest gifts that they provided us was
allowing us to look at two of their
papers concurrently and I think that
next year one of the things that I
really am going to try and um convince
people of is if when not just if but
when we do live streams if people that
have authored papers would be prepared
to come in and show two two or more of
their papers at once because it really
helps us show the potential of looking
through an active inference lens at
work so yeah it was very memorable it
was it was lovely we've we've actually
formed social bonds as a result of the
commitment that they uh have shown to
the active inance community and they
didn't come you know with a with a
background in the active inurance
Community but when we reached out to
them they reached back with open arms it
was really
amazing
yeah all right live stream
54 mathematical foundations for a
compositional account of the basian
brain by Toby St Clair SMI yeah well

first a mega shout out to the whole team
many synchronous and asynchronous
foragers
who met
weekly for all of
2023 with many
incredible persistance and
contributions nonethe least of which
Jr
who absolutely blazed the way
forward with an open source
contribution to a category Theory AC of
inference Adventure story
curriculum
narrative
journey and um the week by week as we
went through this um PhD
dissertation the week by week and the
the journey and the 100 foot waves that
we cited and sometimes got swept away in
just in the months leading up to the
conversation with Toby and then to have
the zero here with Ali but again
building on the contributions of many
people and the months leading up to it
to have the do zero and do the
impossible to bring the dissertation and
the six months into what can you really
say about o infant category Theory which
is like he said what there is to say in
the dissertation so what can we say
about what there was to say so that was
such a fun question and then to have
Toby join and have just such
authenticity and um
relatability with how he approached
category Theory so yeah what would you
ask I was gonna say I think that I think

that the way because I I mean a person
of Toby's intellect that could be able
to write at the level that he wrote for
his dissertation to also be so
approachable is is I guess I guess
normal for him but I would not I would
not have anticipated the approachability
that Toby had the other thing is is that
I I cannot thank Ally enough because the
number of times that we were having our
conversations as we prepared for
this under wave uh the number of times
where I just felt like I I can't be here
I can't be doing this because I got I
got into the first three sections of the
paper and it was and it was quite quite
literally
so it was so unbelievable the amount of
stuff that was packed into this thing I
thought I don't know who the reviewers
of this are but good on
them yeah I I I Echo that thank you Ali
actually showing up to connect the dots
and just kind of be like okay so here's
the thing with the funter we needed that
otherwise we were off the
rails I mean honestly I can I can
appreciate what NASA does now when it
sends something off and it knows that
it's going to meet the asteroid at some
place hundreds of thousands of miles
away and blow it up because that's what
this felt like this felt like I was
trying to find some dot in some distant
land that I had to land on and it was
Ally and and you and and then of course
Toby when he actually showed up and it
was so

chill oh amazing all right live stream
55 realizing synthetic active agents the
triple play in the title and this was a
great Series so first big shout out to
yakob and Bert and also some others but
for helping with a do
zero and for us really coordinating to
prepare this material because this is
kind of another impossible compression
which was to bring together the message
passing on the on the be constrained
free energy on the all of these details
that just seem to stack and stack and
stack and bring us closer and closer to
The Cutting Edge but then
Magnus being so
chill so
clear in the first discussion in the
second discussion showing the code so
sparse so
tur and just being
very friendly and willing to continue to
like help us and learn how to apply
generative models with the RX andur
toolkits and Julia and to make what
literally looks like something like a
circuit diagram or a glass bead game or
some other kind of esoterica what makes
this
have at least a touch point with our
understanding of figure 4.3 or figure
figure 7.3 from the
textbook and just the ways that the
basil motifs of articulation of sense
making and action and policy and
attention all these motifs that we kind
of explore and the base graphs and the
forny graphs that that entails all of

this and then to connect it to the
computational advances like the direct
policy inference so instead of unrolling
policies and then like sampling across
them to do this kind of direct clamped
unclamped direct policy inference on the
generalized free energy it's happening
so fast that it's incredible and it was
just really great that Magnus at all
brought it out so
comprehensively and that
magnus's generosity in also wanting to
see this be really adopted because it's
like a toolkit and a framework that is
being developed in a capacity that's
very fundamental for our
ecosystem not just some Claim about some
system of
Interest a very casual Observer I think
Magnus could translate anything into a
map or a graph anything like he could
translate meditation into a graph or any
and again that's from the Casual
Observer if it again I in my very
colloquial sense say gripper and gripped
and when I was watching you guys go
through this that's exactly what I was
seeing and he was I mean he wasn't using
my language but he was certainly showing
how it can be
done all right and then the last live
stream was
56 so here kind of embodying Dean's
exhortation that authors request a
minimum of two papers Alexander aoria
dispatched four papers to Blue and I so
we
scrambled through to each so thank you

blue for being very generous with with
with working together this this was fun
um and
then Alex joined for the one and the two
we talked about the common model for
cognitive systems and about modeling
diverse intelligences and then in the
second discussion the 56.2 he took it in
the Mortal computation direction that
was like a paper that had out on
pre-print in the time between the one
and the two so it was just incredible
that was like in the
one it was a paper that hasn't come out
yet but oh it'll be cool which just from
like a live stream perspective I'm
always like okay that's cool but by the
time that most people listen to this it
will have been released so it's always
cool to have somebody share that but at
the same time it's kind of like it makes
the work more externally reference IAL
instead of containing the material which
is not a bad thing it's just how that
kind of speech Act is and then for then
the second discussion the prepare
measure with the real world getting in
between the openings and the cracks and
the door is just awesome so this was fun
and uh Alex has a lot to share and it's
just exciting to hear about all these
like relatively very technical detailed
cognitive model architectural motifs
play into bigger topics and
questions
okay all right well here we are
reflections so we have a few different
things here we'll probably write a few

different things down but this is the
last slide so people watching live write
any questions or comments and then we'll
look through it in just one minute but
first in where does this take us or how
does this reflect upon us
now well again if I were to if I were to
say what would
2023 if I was to pick one thing and say
well that that explains 2023 I
think fact that we are talking about
Reflections we're looking back over time
and I think with what Chris was able to
share with us what our many of our
guests were able to share with us the
fundamental nature of how we describe
time how we use time how we prepare and
measure for time what Quantum reference
frames mean in terms of change
Dynamics um I think all of those things
now seem to be a part of the Lexicon
whereas before we would kind of we would
kind of go out look at it a little bit
and then come back to whatever we felt
we were more familiar with
um I don't know what that
means for 2024 other than I think that
if we if we anticipate that we really
still
haven't
um move to a place
where
the the nature of how we
apply active inference perspectives on
the things that we do we're still in the
very nent stages of that I think that I
that the um the way that we incorporate
our understanding of the change and the

Dynamics and the timing not just taking
our time but because I believe that
active inference really affords strategy
how we time our takes is still in front
of us it's still something that I think
we're going to see a a lot of
development in and I think it's going to
be really exciting for people because I
think up until now people are still just
trying to get a grip on what is active
inference they're doing a lot of the
measuring things but I think that when
we actually see the change being and the
and the and the ability to to understand
the change the carrying capacity for
that I think there's going to be a lot
of people that are going to jump on that
perspective and and run with it and are
going to want to really share what that
what that means in terms of the the
things that they do and the hopes that
they want to try and um
realize so that's my sense of what this
year was and it was I think it was quite
a bit different than
2022
yeah yeah 20 22 just from
the top of my head it felt like there
was more developments in basian
mechanics
narrow that was kind of the basian
mechanic um miraculous year of
developments with more paper Focus
discussions yeah heav Kronos in 22 yeah
then we we kept the chronometer up and
also brought in this guest stream the
diversity and people who had um more and
less familiarity with active inference

and higher order discussions about the
role of
technicalities and a formalisms
in philosophies and um the continued
simmering and marinating of the um what
does this mean question multiple kinds
of what does it mean what does this mean
for people what does this mean for
humans what does it mean for sentients
and cognitive systems what does it mean
for the planet what does it mean for all
these different and then
[Music]
um category
Theory painted some incredibly promising
lines
on the
physics that was brought out in
22 we had like the streams and the
videos presented here this isn't a
summary of the ecosystem so a lot of
interesting things happens and we we
only sample arbitrarily truly but one
thing I really look forward to in terms
of like looking back to look forward I
really look forward to working with
Darius and others to integrate the
podcast and live stream into the
production and have an integrated
process for reaching out to people and
scheduling the right kind of event and
the right kind of activity and then just
having a very
professional very Broad and deep and
multilingual
channels and have fun and plug it into
the journal and just like have the
amazing

conversations have the right signaling
and the wisdom to know what is going to
be public and language models and
transcriptions and translations and what
are the conversations that aren't that
way and what's like the bigger picture
also one one kind of image when we were
just before we started and we talked
talked about like how different the
0.1.2 One 2 3 tap tap tap is and about
how the multiple
touches is very different than the kind
of one and through or like one and out
of a guest stream and that kind of one
measurement that appears on the screen
versus that repeated contact and the a
kneeling that happens there so yeah what
do you think about that
well again I I I I you know it's like
your kids you're not supposed to pick
your favorites but I do anyway I but the
way that way that
Chris spent the time with us on the
discussions as much as he presented the
way that the the course of the
interactions in the guest streams where
people had their presentation but they
also made sure that they had time to
actually have that conversation I think
I think that's very powerful and very
valuable I know it's time consuming but
again if we're talking about whether or
not we want to be efficient or whether
we want to be effective um we're just I
think we're really glad that people are
prepared to stick
around and make sure that
that not just for for clarification

purposes I think what I'm finding is is
that people that are now engaging with
us are find in that they're able to
both share and be shared with like it
doesn't feel like it's a
unidirectionality if we have nothing
done nothing else I think we've proven
out that we want to be a community that
we want to create an ethos and that we
want people that want to participate not
just present but
actually get in get in the get in the
ring with us and wrestle with this stuff
because I think as Andreas pointed out
in that last
guest stream
um we take on really hard problems
problems that maybe can't be answered in
in in a 15minute
presentation and I think
that that serves a real purpose so again
if it's if it's about as we grow as an
as an
Institute what we and who what and who
we engage with and
for for what purposes I think at the end
of the today we continue to try and find
as many different ways of organizing
that so that people can come in and like
I said move past the instructional you
know what why that's efficient and move
on to the the uh interactionist because
I think that's where a lot of the stuff
you want to interact with our
library by all means come in and do that
and then come and talk to us and let's
let's find out what value that's that's
serving and how we might be able to

distribute that into a a larger a larger
space or a larger Place nice all right
I'll read some comments from live chat
so upcycle Club thank you Joanne thank
you thank you
Nathaniel all right thank you Susan I'm
going to read your
comment I like what Dean said me too I'm
paraphrasing putting the generative
model to work can we expand on that and
what realistic metrics objectives might
in that regard be for the institute for
2024 aspirational outcomes for 2024 what
will be the challenges both for ACM
Institute and possibilities for using
actim to advance the Evolution for AI in
general I wish how do we how do we think
about challenges I mean do challenges
even exist and actin is something that
helps us address them are challenges
something that are within an active
gripper and gripped relationship with
the work does the otim perspective
reframe what is called a challenge by
other
words well I think one of the things
that we've talked about in you know in a
variety of different spaces is that
there we're going
to we're finding ourselves with with
things like Grant applications being
asked to meet certain standards and I
think we're quite capable of doing that
but I think one of the things that we're
talking aspirationally is
not not just setting ourselves up to
meeting a minimum meeting a standard but
actually trying to cross thresholds and

Boundary cross and move into spaces that
simply are
not being occupied at this point now
that's not we're not colonizing anything
but we are acting as pilgrims and maybe
there are thresholds that we want to get
over and not just barely get over but
maybe really truly kind of break away
from some of the gravitational pull as
long as we can get back right safely but
but that's one of the things that we're
talking about right now not just
necessar meeting standards as that form
of challenge Ares
but also
surpassing maybe by orders of magnitude
so we'll see what's this image mean to
you
here oh I just like things that are
orthagonal
um and looping because I'm a big fan of
learning loops and not necessarily
unidirectional so um whenever I get a
chance to reflect
I can look at things um bidirectionally
and so this now I get the luxury of
being able to look back looking back
isn't linear it's very non-continuous I
can hop around all over the place I can
think of 55.1 and I can also think of uh
the Symposium and certain presenters
there that had the first game the first
animated Game and thinking wow now of
course if I think thought about it in a
linear sense and everything was sort of
sequential that would give me one image
but now that I'm thinking back on it I
get a whole different idea of of just

exactly what that as I say that lap
around the Sun generated for
me yeah in that Spirit here's me in the
passenger
seat looking
forward but taking a picture that
represents back
backwards but backwards coming to the
present all in the forward-looking
moment all captured on the
[Music]
screen all right if anyone has any other
comments I'll just wait like a few more
seconds
otherwise we have a
few I want to thank the officers I want
to thank the board I want to thank the
Sab this you don't have to you don't
have to play me off that's that's it I
don't need to get into some crazy
political statement but again I think
this is a a perfect time to express my
gratitude for all the people that do all
the hard
work thank you I Echo that just speaking
to the
streams and the
videos I really hope if people are
interested in learning these skills of
audiovisual production getting involved
with using some of the automations and
tools that are out there now and helping
like Leverage us to have better and new
kinds of
materials hash con content
creator and Catalyst and just kind of
Enthusiast and um co-learning
all right thank you

Dean farewell thanks everybody

bye